

SECONDARY 3 G3
END-OF-YEAR EXAMINATION 2026

ADDITIONAL MATHEMATICS

4049

Paper 1

2 hours

CANDIDATE NAME: _____

CLASS: 3 _____ INDEX NUMBER: _____

READ THESE INSTRUCTIONS FIRST

Write your full name, class and index number in the spaces above.

Write in dark blue or black pen in the space provided for each question.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer all the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an approved scientific calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 80.

For Examiner's Use					
Q1	4		Q7	7	
Q2	6		Q8	12	
Q3	7		Q9	6	
Q4	8		Q10	6	
Q5	7		Q11	10	
Q6	7		Total	80	
			Total	80	

This document consists of 16 printed pages including the cover page.

Mathematical Formulae

1. ALGEBRA

Quadratic Equation

For the equation $ax^2+bx+c=0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Binomial Expansion

$$(a+b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \dots + \binom{n}{r}a^{n-r}b^r + \dots + b^n$$

where n is a positive integer and

$$\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1)\dots(n-r+1)}{r!}$$

2. TRIGONOMETRY

Identities

$$\sin^2 A + \cos^2 A = 1$$

$$\sec^2 A = 1 + \tan^2 A$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

Formulae for ΔABC

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\Delta = \frac{1}{2}ab \sin C$$

Answer all the questions.

- 1 The line $y = mx - 2$ meets the curve $y = x^2 - 3x + 7$ at two distinct points. Find the range of values of m . [4]

(a) Express $\frac{2\sqrt{3}+5}{3-\sqrt{3}}$ in the form $a+b\sqrt{3}$, where a and b are rational numbers. [3]

(b) Solve the equation $\sqrt{2x+3}=x-1$, giving your answer in exact form. [3]

- 3 It is given that $f(x) = 2x^3 + ax^2 + bx + 6$, where a and b are constants. $(x+2)$ is a factor of $f(x)$, and $f(3) = 0$.

(a) Find the value of a and of b , and hence factorise $f(x)$ completely. [4]

(b) Hence solve the equation $2(2^{3y}) - 3(2^{2y}) - 11(2^y) + 6 = 0$, giving your answers in exact form. [3]

(a) Show that $x^3 - x^2 + 4x - 4 = (x - 1)(x^2 + 4)$. [2]

(b) Hence express $\frac{2x^3 + 2x^2 + 5x + 6}{x^3 - x^2 + 4x - 4}$ in partial fractions. [6]

(a) Solve the equation $\log_3(x+2)+\log_3(x-4)=3$. [3]

(b) Solve the equation $4\log_9 x - \log_x 3 = 1$, giving your answers in exact form. [4]

- 6 The mass, M grams, of a radioactive sample t days after it was first recorded is given by $M = M_0 e^{-kt}$, where M_0 and k are constants. The mass is 500 g when $t = 2$, and 320 g when $t = 10$.

(a) Find the value of k and of M_0 . [4]

(b) Find the least number of complete days, measured from $t = 0$, after which the mass first falls below 100 g. [3]

- (a)** In the expansion of $(2+kx)^6$, where k is a non-zero constant, the coefficient of x^3 is twice the coefficient of x^2 . Find the value of k . [4]

- (b)** Hence find the coefficient of x^2 in the expansion of $(1-2x)(2+3x)^6$. [3]

- 8 The points $A(-2, 3)$ and $B(6, 7)$ are joined by a straight line. Solutions by accurate drawing will not be accepted.

(a) Find the equation of the perpendicular bisector of AB . [3]

(b) The perpendicular bisector of AB cuts the x -axis at the point C . Find the coordinates of C . [2]

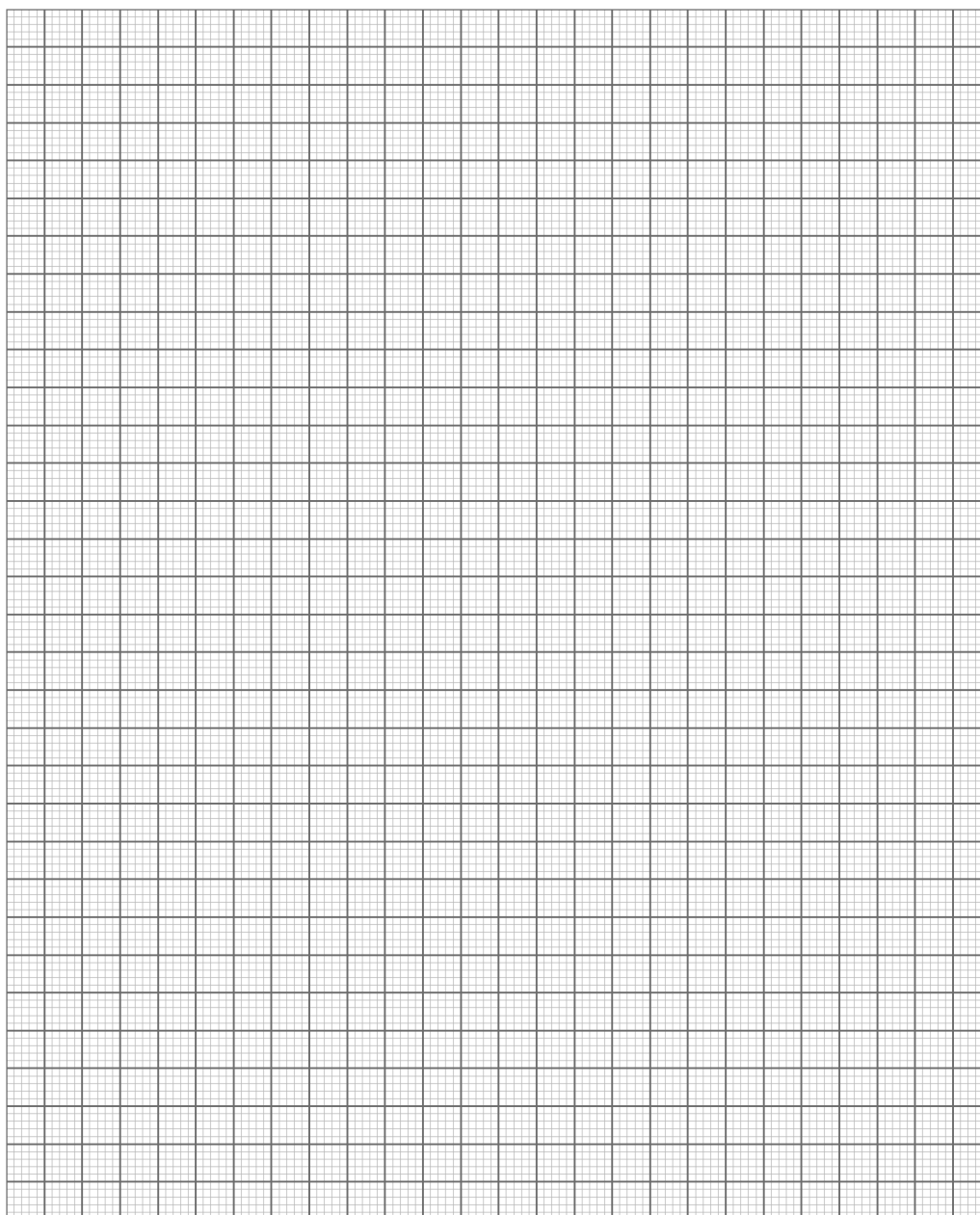
(c) Explain why triangle ABC is isosceles. [2]

(d) The point D lies on the perpendicular bisector of AB and the area of triangle ABD is 30 units². Find the possible coordinates of D . [5]

- 9 The variables x and y are related by the equation $y = ax^b$, where a and b are constants. The table shows measured values of x and y .

x	1	4	16	36	100
y	3.00	6.00	12.0	18.0	30.0

- (a) Explain how a straight line graph may be drawn to represent this equation, and draw it for the given data on the grid below. [3]



(b) Use your graph to estimate the value of a and of b .

[3]

10 Given that $\sin C = q$, where C is reflex and $180^\circ < C < 270^\circ$, express each of the following in terms of q .

(i) $\cos C$ [2]

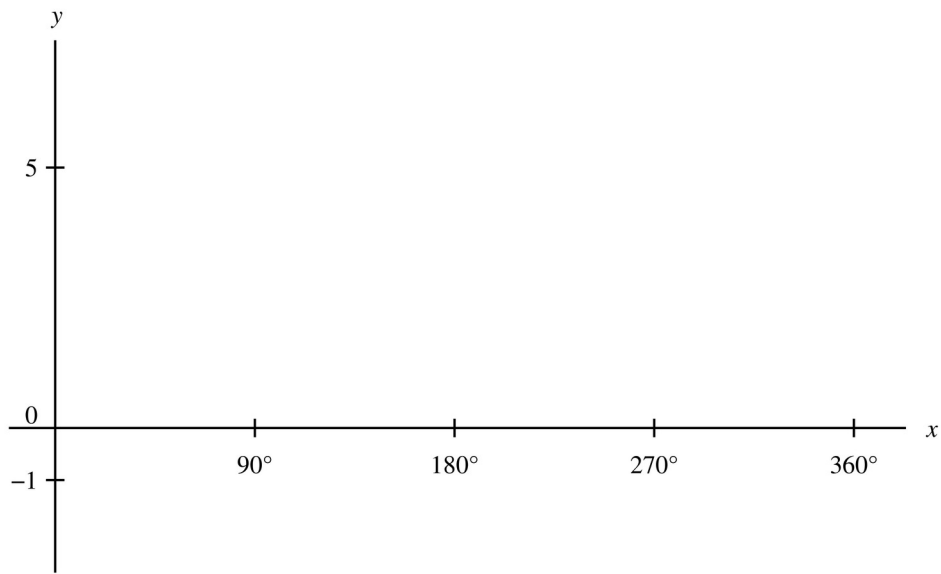
(ii) $\tan C$ [2]

(iii) $\sec(360^\circ - C)$ [2]

(a) Prove the identity $\frac{1-\cos \theta}{\sin \theta} + \frac{\sin \theta}{1-\cos \theta} = 2 \operatorname{cosec} \theta$. [3]

(b) Hence solve the equation $\frac{1-\cos \theta}{\sin \theta} + \frac{\sin \theta}{1-\cos \theta} = 4$ for $0^\circ \leq \theta \leq 360^\circ$. [3]

- (c) Sketch the graph of $y=2+3\sin 2x$ for $0^\circ \leq x \leq 360^\circ$, stating the amplitude and the period of the curve. [4]



----- End of Paper -----